conf_matrix = confusion_matrix(y_test, predictions) print("Confusion Matrix:\n", conf_matrix)

ROC Curve and AUC

The ROC curve illustrates the trade-off between true positive rate (recall) and false positive rate at various thresholds. The Area Under the Curve (AUC) summarizes the model's overall ability to distinguish between classes: from sklearn.metrics import roc_curve, auc

import matplotlib.pyplot as plt

Example usage
fpr, tpr, = *roc*curve(y_test, predictions_proba) roc_auc = auc(fpr, tpr)

plt.plot(fpr, tpr, label=f"ROC curve (area = {roc_auc:.2f})") plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend(loc="lower right")
plt.show()

Evaluating a model's performance requires careful consideration of the task at hand and the dataset's characteristics. While accuracy is a good starting point, metrics like precision, recall, and F1-score provide deeper insights, especially for imbalanced datasets. By using these metrics effectively, you can ensure that your models perform reliably and meet the specific requirements of your application.

Real-World Example: Predicting House Prices

Predicting house prices is a classic regression problem that leverages supervised learning techniques. It involves estimating the price of a house based on various features such as square footage, number of bedrooms, and location. This example demonstrates the power of machine learning in solving real-world problems and highlights the importance of combining domain knowledge with technical expertise.

Why Predict House Prices?

House price prediction is a valuable application of machine learning for:

- **Real Estate Professionals:** Assisting in pricing properties competitively.
- **Homebuyers:** Helping buyers assess whether a property is priced fairly.